1.30 Differentiation from First Principles

Worked Solutions

1a)

$$f(1) = 2 + 1 = 3.$$

1b)

$$f(-1) = 2 + 1 = 3.$$

1c)

$$f(3) = 18 + 1 = 19.$$

1d)

$$f(x) = 32 + 1 = 33.$$

1e)

$$f\left(\frac{1}{2}\right) = \frac{1}{2} + 1 = \frac{3}{2}.$$

2a)

$(0, 1)$ and $(1, 5)$ so $\dfrac{5-1}{1-0} = 4.$

2b)

$(0, 2)$ and $(1, 3)$ so $\dfrac{3-2}{1-0} = 1.$

2c)

$\left(0, -\dfrac{1}{2}\right)$ and $(1, -1)$ so $\dfrac{-1+1/2}{1-0} = -\dfrac{1}{2}.$

2d)

$(0, 0)$ and $\left(1, \dfrac{1}{2}\right)$ so $\dfrac{1/2-0}{1-0} = \dfrac{1}{2}.$

3)

28.028007, 2.0001, 28.00280007.

Hence 28.

4a)

2, 1.43547, 1.39111, 1.38678, 1.38634.

4b)

1.39.

5a)

$(2, 9.2)$ and $(3, 9.68)$ so $\dfrac{9.68 - 9.2}{3 - 2} = 0.48$.

5b)

$(2.1, 9.270045171\ldots)$
so $\dfrac{9.270045171 - 9.2}{2.1 - 2} = 0.7004517076 = 0.700$ to 3sf.

5c)

Use a third point that is on the curve which is even closer to point P, so that its x-coordinate is $2 < x < 2.1$.

6a)

$(2.5, -2)$ so $\dfrac{-2 + 1}{2.5 - 2} = -2$.

6b)

At $x = 3$ the function is undefined (discontinuous).

6c)

$(3.5, 2)$ so $\dfrac{2 + 1}{3.5 - 2} = 2$ however there is a discontinuity between the two points, so it does not approximate the gradient at $x = 2$.

7a)

$$1 + 2h + h^2$$

7b)

$$5 + 10h + 5h^2$$

7c)

$$1 + 3h + 3h^2 + h^3$$

7d)

$$-3 - 9h - 9h^2 - 3h^3$$

8a)

$$4 + 4h + h^2$$

8b)

$$7(9 + 6h + h^2) = 63 + 42h + 7h^2$$

8c)

$$(-2)^3 + 3 \times (-2)^2 h + 3 \times (-2)h^2 + h^3 = -8 + 12h - 6h^2 + h^3$$

8d)

$$-3(10^3 + 3 \times 10^2 h + 3 \times 10h^2 + h^3) = -3000 - 900h - 90h^2 - 3h^3$$

9a)

$$9 + 6h + h^2 - 9 = 6h + h^2$$

9b)

$$6(1 + 2h + h^2) - 6 = 12h + 6h^2$$

9c)

$$(-1)^3 + 3 \times (-1)^2 h + 3 \times (-1)h^2 + h^3 + 1 = 3h - 3h^2 + h^3$$

9d)

$$-2(4^3 + 3 \times 4^2 h + 3 \times 4h^2 + h^3) + 128 = -96h - 24h^2 - 2h^3$$

10a)

$$f(2) = 4, \qquad f(2 + h) = 4 + 4h + h^2.$$

10b)

$$f(2) = 2, \qquad f(2 + h) = 2 + 3h + h^2.$$

10c)

$$f(2) = 5, \qquad f(2+h) = 5 + 2h + h^2.$$

10d)

$$f(2) = -10, \qquad f(2+h) = -10 - 11h - 3h^2.$$

10e)

$$f(2) = 0, \qquad f(2+h) = -9h - 2h^2.$$

11a)

2.

11b)

-5.

11c)

3.

11d)

$$\lim_{h \to 0}(5h + 1) = 1.$$

11e)

$$\lim_{h \to 0}(6h - 4) = -4.$$

12)

$\dfrac{h^2 - 2}{h} = 1 - \dfrac{2}{h}$, which is undefined when $h = 0$.

As $h \to 0$, $\dfrac{2}{h}$ diverges.

13a)

$$1 + 3h + 3h^2 + h^3$$

13b)

$$\frac{8(1+h)^3 - 8}{h} = \frac{24h + 24h^2 + 8h^3}{h} = 24 + 24h + 8h^2.$$

13c)

$$m = \lim_{h\to 0}(24 + 24h + 8h^2) = 24.$$

14a)

$$f(2) = 4, \quad f(2+h) = 4 + 4h + h^2.$$
$$\frac{dy}{dx} = \lim_{h\to 0}\frac{4 + 4h + h^2 - 4}{h} = \lim_{h\to 0}\frac{4h + h^2}{h} = \lim_{h\to 0}(4 + h) = 4.$$

14b)

$$f(1) = 3, \quad f(1+h) = 3 + 6h + 3h^2.$$
$$\frac{dy}{dx} = \lim_{h\to 0}\frac{3 + 6h + 3h^2 - 3}{h} = \lim_{h\to 0}\frac{6h + 3h^2}{h} = \lim_{h\to 0}(6 + 3h) = 6.$$

14c)

$$f(-1) = 6, \quad f(-1+h) = 1 - 2h + h^2 - 5(-1+h) = 6 - 7h + h^2.$$
$$\frac{dy}{dx} = \lim_{h\to 0}\frac{6 - 7h + h^2 - 6}{h} = \lim_{h\to 0}\frac{-7h + h^2}{h} = \lim_{h\to 0}(-7 + h) = -7.$$

15a)

$(4, 21)$, $(4.1, 21.81)$ so $\dfrac{21.81 - 21}{4.1 - 4} = 8.1.$

15b)

$$f(4+h) = 16 + 8h + h^2 + 5 = 21 + 8h + h^2.$$
$$\frac{21 + 8h + h^2 - 21}{h} = \frac{8h + h^2}{h} = 8 + h.$$

15c)

8.

16a)

$$(3+h, 5(3+h)^2) = (3+h, 45 + 30h + 5h^2).$$
$$\frac{45 + 30h + 5h^2 - 45}{3 + h - 3} = \frac{30h + 5h^2}{h} = 30 + 5h.$$
$$\therefore m = \lim_{h\to 0}(30 + 5h).$$

16b)

$m = 30$.

17)

$$f(x) = 5x^2, \quad f(x+h) = 5x^2 + 10xh + 5h^2.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{5x^2 + 10xh + 5h^2 - 5x^2}{h} = \lim_{h\to 0} \frac{10xh + 5h^2}{h}.$$
$$= \lim_{h\to 0}(10x + 5h) = 10x.$$

18a)

$$f(x) = 2x^2 + 1, \quad f(x+h) = 2x^2 + 4xh + 2h^2 + 1.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{2x^2 + 4xh + 2h^2 + 1 - (2x^2 + 1)}{h} = \lim_{h\to 0} \frac{4xh + 2h^2}{h}.$$
$$= \lim_{h\to 0}(4x + 2h) = 4x.$$

18b)

$$f(x) = 5x^2 + x, \quad f(x+h) = 5x^2 + 10xh + 5h^2 + x + h.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{5x^2 + 10xh + 5h^2 + x + h - (5x^2 + x)}{h}.$$
$$= \lim_{h\to 0} \frac{10xh + 5h^2 + h}{h} = \lim_{h\to 0}(10x + 5h + 1) = 10x + 1.$$

18c)

$$f(x) = x - 4x^2, \quad f(x+h) = x + h - 4x^2 - 8xh - 4h^2.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{x + h - 4x^2 - 8xh - 4h^2 - (x - 4x^2)}{h}.$$
$$= \lim_{h\to 0} \frac{h - 8xh - 4h^2}{h} = \lim_{h\to 0}(1 - 8x - 4h) = 1 - 8x.$$

18d)

$$f(x) = x^2 - 2x + 1, \quad f(x+h) = x^2 + 2xh + h^2 - 2x - 2h + 1.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 2x - 2h + 1 - (x^2 - 2x + 1)}{h}.$$
$$= \lim_{h\to 0} \frac{2xh + h^2 - 2h}{h} = \lim_{h\to 0}(2x + h - 2) = 2x - 2.$$

18e)

$$f(x) = x^3, \quad f(x+h) = x^3 + 3x^2h + 3xh^2 + h^3.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3}{h}.$$
$$= \lim_{h\to 0}(3x^2 + 3xh + h^2) = 3x^2.$$

19a)

$$f(2) = 12, \quad f(2+h) = 3(4 + 4h + h^2) = 12 + 12h + 3h^2.$$
$$f'(2) = \lim_{h\to 0} \frac{12 + 12h + 3h^2 - 12}{h} = \lim_{h\to 0} \frac{12h + 3h^2}{h} = \lim_{h\to 0}(12 + 3h) = 12.$$

19b)

$$f(2) = -7, \quad f(2+h) = 1 - 2(4 + 4h + h^2) = -7 - 8h - 2h^2.$$
$$f'(2) = \lim_{h\to 0} \frac{-7 - 8h - 2h^2 + 7}{h} = \lim_{h\to 0} \frac{-8h - 2h^2}{h} = \lim_{h\to 0}(-8 - 2h) = -8.$$

20)

$$f(x) = kx^3, \quad f(x+h) = kx^3 + 3kx^2h + 3kxh^2 + kh^3.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{kx^3 + 3kx^2h + 3kxh^2 + kh^3 - kx^3}{h}.$$
$$= \lim_{h\to 0} \frac{3kx^2h + 3kxh^2 + kh^3}{h} = \lim_{h\to 0}(3kx^2 + 3kxh + kh^2) = 3kx^2.$$

21a)

$$f(2) = 16, \quad f(2+h) = 12(2+h) - (2+h)^3.$$
$$= 24 + 12h - (8 + 12h + 6h^2 + h^3) = 16 - 6h^2 - h^3.$$

So $\dfrac{16 - 6h^2 - h^3 - 16}{2 + h - 2} = \dfrac{-6h^2 - h^3}{h} = -6h - h^2.$

21b)

$$\lim_{h\to 0}(-6h - h^2) = 0.$$

22)

The expansion of $2(3+h)^3$ and missing $\lim_{h\to 0}$ on penultimate line.

$$f(3) = 2 \times 3^3 - 9 = 45.$$
$$f(3+h) = 2(3+h)^3 - 9 = 2(27 + 27h + 9h^2 + h^3) - 9.$$
$$= 2h^3 + 18h^2 + 54h + 45.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{2h^3 + 18h^2 + 54h + 45 - 45}{h} = \lim_{h\to 0}(2h^2 + 18h + 54) = 54.$$

23)

$$f(x) = ax + b, \quad f(x+h) = ax + ah + b.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{ax + ah + b - (ax + b)}{h} = \lim_{h\to 0} \frac{ah}{h} = \lim_{h\to 0}(a) = a.$$

24)

$$f(x) = ax^2 + bx + c, \quad f(x+h) = ax^2 + 2axh + ah^2 + bx + bh + c.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{ax^2 + 2axh + ah^2 + bx + bh + c - (ax^2 + bx + c)}{h}.$$

$$= \lim_{h\to 0} \frac{2axh + ah^2 + bh}{h} = \lim_{h\to 0}(2ax + ah + b) = 2ax + b.$$

25)

$(3, 8)$ and $(a, 16)$.

$16 = a^2 + 2a - 7$ gives $a = -1 \pm 2\sqrt{6}$ but $a > 3$, so $a = -1 + 2\sqrt{6}$.

Note that $(a, 0) \Rightarrow a < 3$.

26)

$$f(2) = 8p, \quad f(2+h) = 8p + 12ph + 6ph^2 + ph^3.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{8p + 12ph + 6ph^2 + ph^3 - 8p}{h}.$$

$$= \lim_{h\to 0} \frac{12ph + 6ph^2 + ph^3}{h} = \lim_{h\to 0}(12p + 6ph + ph^2) = 12p.$$

$12p = 24$ so $p = 2$.

$$g(2) = 4q - 20 + 1 = 4q - 19, \quad g(x+h) = 4q + 4qh + qh^2 - 10h - 19.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{4q + 4qh + qh^2 - 10h - 19 - (4q - 19)}{h}.$$

$$= \lim_{h\to 0} \frac{4qh + qh^2 - 10h}{h} = \lim_{h\to 0}(4q + qh - 10) = 4q - 10.$$

$4q - 10 = 24$ so $q = \dfrac{17}{2}$.

27a)

$$f(2) = 9, \quad f(2+h) = 16 + 16h + 4h^2 - 7 = 9 + 16h + 4h^2.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{9 + 16h + 4h^2 - 9}{h} = \lim_{h\to 0} \frac{16h + 4h^2}{h} = \lim_{h\to 0}(16 + 4h) = 16.$$

27b)

$$\arctan\left(\frac{16}{1}\right) = 86.4^\circ.$$

28a)

$$f(1) = 4, \quad f(1+h) = 4(1+h)^2 - (1+h) + 1.$$
$$= 4 + 8h + 4h^2 - 1 - h + 1 = 4h^2 + 7h + 4.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{4h^2 + 7h + 4 - 4}{h} = \lim_{h\to 0} \frac{4h^2 + 7h}{h} = \lim_{h\to 0}(4h + 7) = 7.$$

28b)

$$f(x) = 4x^2 - x + 1, \quad f(x+h) = 4(x+h)^2 - (x+h) + 1.$$
$$= 4x^2 + 8xh + 4h^2 - x - h + 1.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{4x^2 + 8xh + 4h^2 - x - h + 1 - (4x^2 - x + 1)}{h}.$$
$$= \lim_{h\to 0} \frac{8xh + 4h^2 - h}{h} = \lim_{h\to 0}(8x + 4h - 1) = 8x - 1.$$

$8x - 1 = 15$ so at the point $x = 2$.

29a)

$$f(x) = \frac{1}{x}, \quad f(x+h) = \frac{1}{x+h}.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h\to 0} \frac{\frac{x-(x+h)}{(x+h)x}}{h} = \lim_{h\to 0} \frac{x-(x+h)}{h(x+h)x}.$$
$$= \lim_{h\to 0} \frac{-h}{h(x+h)x} = \lim_{h\to 0} \frac{-1}{(x+h)x} = -\frac{1}{x^2}.$$

29b)

$$f(x) = \frac{1}{1-x}, \quad f(x+h) = \frac{1}{1-(x+h)}.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\frac{1}{1-(x+h)} - \frac{1}{1-x}}{h} = \lim_{h\to 0} \frac{\frac{(1-x)-(1-(x+h))}{(1-(x+h))(1-x)}}{h}.$$
$$= \lim_{h\to 0} \frac{(1-x)-(1-(x+h))}{h(1-(x+h))(1-x)} = \lim_{h\to 0} \frac{h}{h(1-(x+h))(1-x)}.$$
$$= \lim_{h\to 0} \frac{1}{(1-(x+h))(1-x)} = \frac{1}{(1-x)^2}.$$

29c)

$$f(x) = \sqrt{x+1}, \quad f(x+h) = \sqrt{x+h+1}.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\sqrt{x+h+1} - \sqrt{x+1}}{h}.$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h+1} - \sqrt{x+1}}{h} \times \frac{\sqrt{x+h+1} + \sqrt{x+1}}{\sqrt{x+h+1} + \sqrt{x+1}}.$$

$$= \lim_{h\to 0} \frac{x+h+1-(x+1)}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \lim_{h\to 0} \frac{h}{h(\sqrt{x+h+1} + \sqrt{x+1})}.$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{x+h+1} + \sqrt{x+1}} = \frac{1}{\sqrt{x+1} + \sqrt{x+1}} = \frac{1}{2\sqrt{x+1}}.$$

30)

$$f(x) = \frac{1}{7x+2}, \quad f(x+h) = \frac{1}{7(x+h)+2}.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\frac{1}{7(x+h)+2} - \frac{1}{7x+2}}{h} = \lim_{h\to 0} \frac{\frac{(7x+2)-(7(x+h)+2)}{(7(x+h)+2)(7x+2)}}{h}.$$

$$= \lim_{h\to 0} \frac{(7x+2)-(7(x+h)+2)}{h(7(x+h)+2)(7x+2)} = \lim_{h\to 0} \frac{-7h}{h(7(x+h)+2)(7x+2)}.$$

$$= \lim_{h\to 0} \frac{-7}{(7(x+h)+2)(7x+2)} = -\frac{7}{(7x+2)^2}.$$

31)

$$f(x) = \frac{2}{3x-1}, \quad f(x+h) = \frac{2}{3(x+h)-1}.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\frac{2}{3(x+h)-1} - \frac{2}{3x-1}}{h} = \lim_{h\to 0} \frac{\frac{2(3x-1)-2(3(x+h)-1)}{(3(x+h)-1)(3x-1)}}{h}.$$

$$= \lim_{h\to 0} \frac{2(3x-1)-2(3(x+h)-1)}{h(3(x+h)-1)(3x-1)} = \lim_{h\to 0} \frac{-6h}{h(3(x+h)-1)(3x-1)}.$$

$$= \lim_{h\to 0} \frac{-6}{(3(x+h)-1)(3x-1)} = -\frac{6}{(3x-1)^2}.$$

32a)

$$f(2) = 8 - 24 = -16, \quad f(2+h) = (2+h)^3 - 12(2+h).$$

$$= 8 + 12h + 6h^2 + h^3 - 24 - 12h = h^3 + 6h^2 - 16.$$

$$\frac{h^3 + 6h^2 - 16 + 16}{h} = \frac{h^3 + 6h^2}{h} = h^2 + 6h.$$

32b)

As $h \to 0$, $h^2 + 6h \to 0$ so the gradient is zero, hence a stationary point.

33a)

$$f(x) = x^5, \quad f(x+h) = x^5 + 5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{x^5 + 5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5 - x^5}{h}.$$
$$= \lim_{h\to 0} \frac{5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5}{h}.$$
$$= \lim_{h\to 0}(5x^4 + 10x^3h + 10x^2h^2 + 5xh^3 + h^4) = 5x^4.$$

33b)

$$f(x) = 2x^4, \quad f(x+h) = 2x^4 + 8x^3h + 12x^2h^2 + 8xh^3 + h^4.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{2x^4 + 8x^3h + 12x^2h^2 + 8xh^3 + h^4 - 2x^4}{h}.$$
$$= \lim_{h\to 0} \frac{8x^3h + 12x^2h^2 + 8xh^3 + h^4}{h} = \lim_{h\to 0}(8x^3 + 12x^2h + 8xh^2 + h^3) = 8x^3.$$

34a)

$$f(x) = (1-x)^3 = 1 - 3x + 3x^2 - x^3.$$
$$f(x+h) = (1-(x+h))^3 = 1 - 3(x+h) + 3(x+h)^2 - (x+h)^3.$$
$$= 1 - 3(x+h) + 3(x^2 + 2xh + h^2) - (x^3 + 3x^2h + 3xh^2 + h^3).$$
$$= 1 - 3x - 3h + 3x^2 + 6xh + 3h^2 - x^3 - 3x^2h - 3xh^2 - h^3.$$
$$-x^3 + 6x^2 - 12x + 8f(x+h) - f(x) = 1 - 3x - 3h + 3x^2 + 6xh + 3h^2 - x^3 - 3x^2h -$$
$$3xh^2 - h^3 - (1 - 3x + 3x^2 - x^3)$$
$$= -3h + 6xh + 3h^2 - 3x^2h - 3xh^2 - h^3.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{-3h + 6xh + 3h^2 - 3x^2h - 3xh^2 - h^3}{h}.$$
$$= \lim_{h\to 0}(-3 + 6x + 3h - 3x^2 - 3xh - h^2).$$
$$= -3 + 6x - 3x^2 = -3(x^2 - 2x + 1) = -3(1-x)^2.$$

34b)

$$f(x) = (2-x)^3 = 8 - 12x + 6x^2 - x^3.$$
$$f(x+h) = (2-(x+h))^3 = 8 - 12(x+h) + 6(x+h)^2 - (x+h)^3.$$
$$= 8 - 12(x+h) + 6(x^2 + 2xh + h^2) - (x^3 + 3x^2h + 3xh^2 + h^3).$$
$$= 8 - 12x - 12h + 6x^2 + 12xh + 6h^2 - x^3 - 3x^2h - 3xh^2 - h^3.$$
$$f(x+h) - f(x) = 8 - 12x - 12h + 6x^2 + 12xh + 6h^2 - x^3 - 3x^2h - 3xh^2 - h^3 -$$
$$(8 - 12x + 6x^2 - x^3) = -12h + 12xh + 6h^2 - 3x^2h - 3xh^2 - h^3.$$
$$\frac{dy}{dx} = \lim_{h\to 0} \frac{-12h + 12xh + 6h^2 - 3x^2h - 3xh^2 - h^3}{h}.$$
$$= \lim_{h\to 0}(-12 + 12x + 6h - 3x^2 - 3xh - h^2).$$
$$= -12 + 12x - 3x^2 = -3(x^2 + 4x - 4) = -3(2-x)^2.$$

34c)

$$f(x) = (1-2x)^3 = 1 - 6x + 12x^2 - 8x^3.$$

$$f(x+h) = (1-(x+h))^3 = 1 - 6(x+h) + 12(x+h)^2 - 8(x+h)^3.$$

$$= 1 - 6(x+h) + 12(x^2 + 2xh + h^2) - 8(x^3 + 3x^2h + 3xh^2 + h^3).$$

$$= 1 - 6x - 6h + 12x^2 + 24xh + 12h^2 - 8x^3 - 24x^2h - 24xh^2 - 8h^3.$$

$$1 - 6x - 6h + 12x^2 + 24xh + 12h^2 - 8x^3 - 24x^2h - 24xh^2 - 8h^3 -$$

$$(1 - 6x + 12x^2 - 8x^3)$$

$$= -6h + 24xh + 12h^2 - 24x^2h - 24xh^2 - 8h^3.$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{-6h + 24xh + 12h^2 - 24x^2h - 24xh^2 - 8h^3}{h}.$$

$$= \lim_{h\to 0} (-6 + 24x + 12h - 24x^2 - 24xh - 8h^2).$$

$$= -6 + 24x - 24x^2 = -6(4x^2 - 4x + 1) = -6(1-2x)^2.$$